

PROJECT PRESENTATION

Ticket Resolution Multi-Agent-System

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AGENDA

1. The business, the data, the problem
2. Solution
3. Data
4. Request Flow
5. Multi-Agent Architecture
6. Trusting the AI
7. My approach
8. Challenge
9. Decisions
10. Questions

THE BUSINESS AND THE PROBLEM

- LUMA Wellness provides services in the wellness and beauty industry.
- Its customer support platform manages complaints related to appointments, payments, memberships, and bookings.
- Each complaint traditionally requires manual investigation by support staff across multiple systems.

PROPOSED SOLUTION

- Build an AI-assisted system that can investigate customer complaints and recommend appropriate resolutions.
- Use a multi-agent workflow to manage the process from complaint intake through investigation, policy review, recommendation, and decision.
- Automatically resolve routine, low-risk cases when sufficient evidence is available.
- Require human review and approval for irreversible or financial actions, such as issuing refunds.
- Provide support staff with research tools to inspect evidence, review policies, and investigate cases directly within the platform.

DATA

We would be primarily dealing with two kinds of data

Operations Data

- Structured business records used by the agents to investigate customer complaints and establish evidence.
- Details like Customer info, memberships, appointments etc
- Stored in a persistent SQL database

Policy Data

- Business rules used to determine the appropriate resolution for each complaint.
- Uses the policy version effective when the incident occurred.
- Stored in vector database for semantic retrieval

DASHBOARD OVERVIEW

Luma
RESOLUTION CENTER

Overview

Case queue

Approvals

Business records

Policy search

Luma Operations
Operations

Sign out

OPERATIONS WORKSPACE

Evidence-grounded case resolution

LIVE

Good Morning.

Here's how the resolution queue is moving.

View all cases

Active cases

10

Queued or processing

Pending approval

10

Waiting for your decision

Human investigation

37

Evidence or policy gaps

Resolution rate

41%

Avg. resolution 170.4s

LATEST ACTIVITY

Case queue

CASE	CATEGORY	STATUS	SOURCE	AGE
CASE-34B4155BFCEC I never received a confirmation after the booking page crashed for ...	Missing Appointment	Human Investigation	simulator	24m
CASE-87E10026AEE9 Could you review the \$60 cancellation fee from my Chicago appoint...	Cancellation Fee Dispute	Resolved	simulator	25m
CASE-165AF6B6E83D Why couldn't I book a noon massage at the Denver location for Mon...	Online Booking Unavailable	Resolved	simulator	25m
CASE-0AAA305EB6EF There's a \$60 payment still pending on my card from Luma. Do I ne...	Duplicate Payment	Resolved	simulator	25m
CASE-B98F3190EBE4 I received a promotional gift card, but the code keeps coming up as i...	Unknown	Human Investigation	simulator	25m
CASE-40BE40D0F6DB I was charged \$120 twice for the same massage in Indianapolis on J...	Duplicate Payment	Pending Approval	simulator	26m

CUSTOMER REQUEST INTERFACE

CUSTOMER CARE

How can we help?

Tell us what happened. We'll review your account activity and the policy that applies.

Customer reference

e.g. CUS-0001

Contact email

you@example.com

What do you need help with?

Select an issue



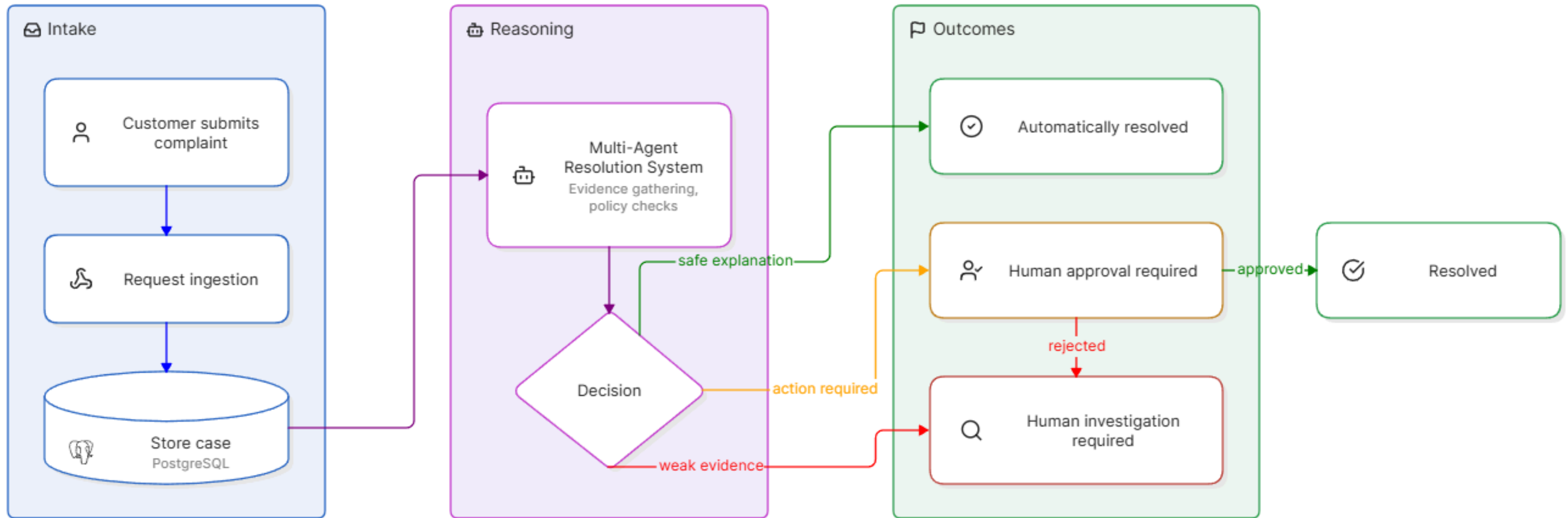
What happened?

Share useful dates, locations, or references if you have them.



Submit case

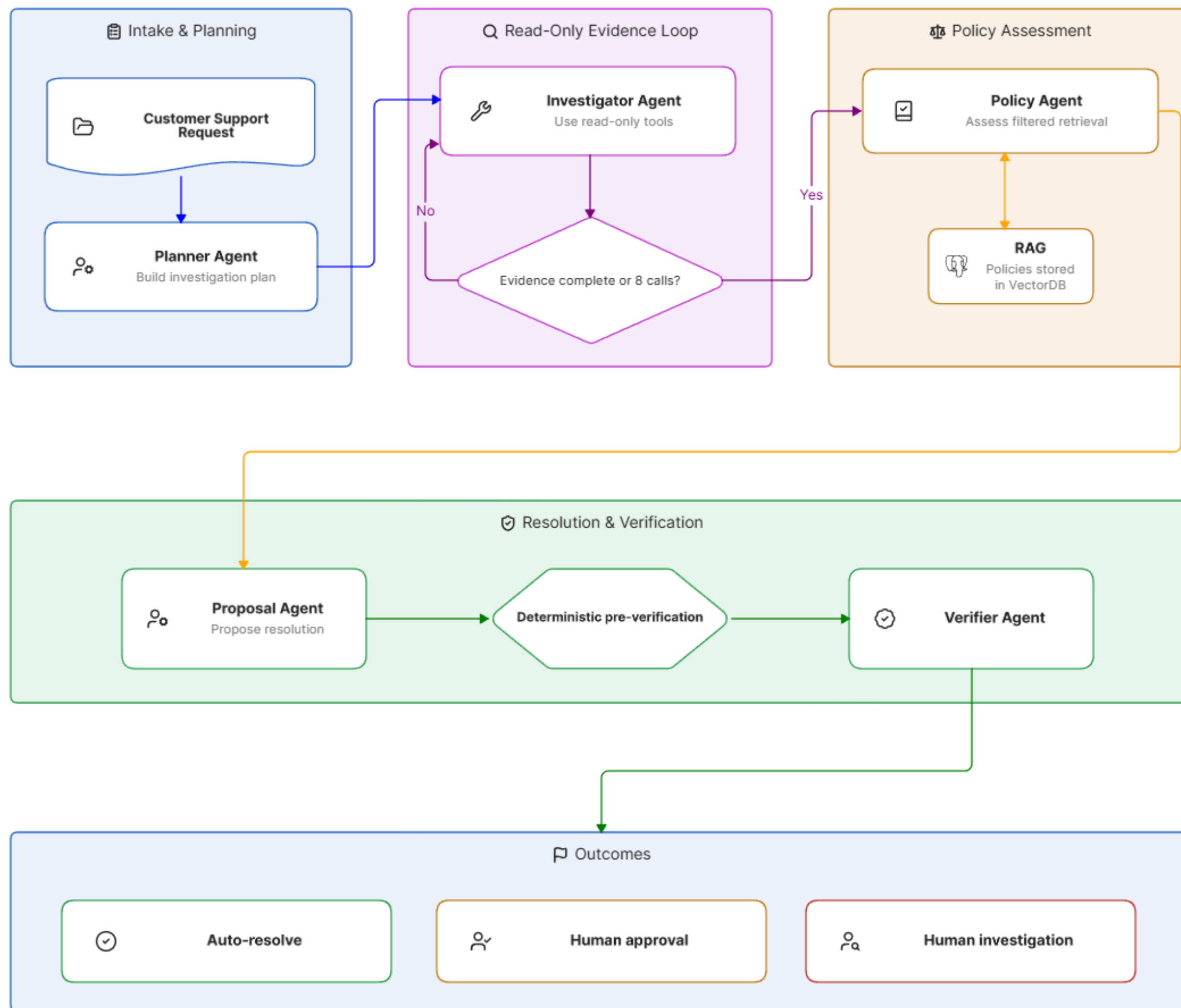
REQUEST FLOW



THE AI ARCHITECTURE

- 5 specialized AI Agents
- Structured LLM Outputs
- Human in the loop approval
- 3 controlled outcomes

Case Resolution Agent Workflow



HOW AGENTS COMMUNICATE

</> JSON



```
{
  "case_id": "case-uuid",
  "case_reference": "CASE-8A42F190",
  "case_run_id": "71eeb913-1471-40ac-9ca5-a9f45f55f012",
  "case_received_at": "2026-09-08T10:15:00Z",
  "complaint_text": "The provider cancelled my appointment, but I was still charged an $80 cancellation fee.",
  "customer_reference": "CUS-0142",
  "claimed_category": "cancellation_fee_dispute",
  "evidence": [],
  "escalation_reason": null
}
```

Information added as the investigation progresses:

- plan
- evidence_coverage
- policy_result
- policy_assessment
- proposal
- verification
- final_disposition
- approval_decision

EXAMPLE

CASE-47891B02C92B

Cancellation Fee Dispute

CUS-0004 · simulated.customer@example.test · 7/9/2026, 4:59:30 pm

Resolved

Research customer records

CUSTOMER COMPLAINT

“I had to cancel my July 11 noon massage a few hours before it started, and now I see a \$60 charge. Can you explain why it was applied?”

RECOMMENDATION

Keep the \$60 cancellation fee

VERIFIED

The customer cancelled the appointment 4 hours before it was scheduled to begin. Luma requires at least 24 hours' notice to avoid a cancellation fee. The \$60 charge is allowed under the policy.

Cancelled by customer

4 hours before it was scheduled to begin

\$60 charged

Next step No further action is required.

► [View decision details](#)

SUPPORTING RECORDS

Evidence reviewed

4

✓ **Customer Identity**
Matthew Wagner · identity and contact status checked

USED IN DECISION [View details](#)

✓ **Booking History**
1 record reviewed

USED IN DECISION [View details](#)

✓ **Appointment Timeline**
APPT-CAN-LATE · Cancelled · 2 timeline events

USED IN DECISION [View details](#)

✓ **Payment**
1 payment across 1 invoice · \$60.00 captured

USED IN DECISION [View details](#)

CASE HISTORY

What happened

- **Case Received**
4:59:30 Pm · System
- **Added To Review Queue**
4:59:30 Pm · System
- **Investigation Started**
5:01:25 Pm · System
- **Case Resolved**
5:02:24 Pm · System
- **Customer Email Draft Prepared**
5:02:24 Pm · System

EXAMPLE

KNOWLEDGE

2

Policies applied

Notice interval

Cancellation and No-Show Policy · POL-CAN-v2#2.1

[Read policy](#)

Standard notice

Cancellation and No-Show Policy · POL-CAN-v2#3.1

[Read policy](#)

CUSTOMER NOTIFICATION

REVIEW REQUIRED

Final email draft

To simulated.customer@example.test

Subject

Resolution for your Luma Wellness case CASE-47891B02C92B

Message

Hello,

We reviewed the \$60 cancellation fee for your July 11, 2026 appointment. Our records show that you cancelled your July 11, 2026 appointment 4 hours before it was scheduled to begin. Luma requires at least 24 hours' notice to avoid a cancellation fee. Based on the verified records and applicable policy, the fee was correctly charged and no refund is due.

Case reference: CASE-47891B02C92B

Luma Wellness Support

Prototype mode: Send email records a mock delivery; no external email is sent.

Save draft

Send Email

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TRUSTING THE AI

To make the system safe, reliable, and trustworthy, I focused on two key areas:

1. Guardrails - Define what the AI can do and when a human must intervene.
2. Evaluations - Measuring whether the system makes accurate, evidence-backed, and safe decisions.

GUARDRAILS

- Structured Model Outputs - validated through pydantic schema checks
- Deterministic Pre-verification - checks for evidence and policy citations
- A separate Verification agent - acting as LLM as a judge
- Human approval gate - the agents don't have access to write tools.

EVALS

How It Works

- Expected results are defined for each case
- Agent outputs are graded automatically
- Both the final answer and decision path are checked
- Held-out cases test generalization

What We Evaluate

- Final decision and outcome
- Evidence gathered
- Policies retrieved
- Tools selected
- Action detail

Evaluation Dataset

60 evaluation cases - 60% development cases and 40% held-out cases

RESULTS

Quality • Classification Accuracy: 91% • Decision Accuracy: 79% • RAG recall@5: 94.12%	Safety • Unauthorized action rate: 0% • Duplicate execution rate: 0% • Structured-output success: 90% (9/10)
Latency • End-to-end P50: 59.2 seconds • End-to-end P95: 255.7 seconds	Cost (per complaint) • Input: 43,644 tokens • Output: 2,104 tokens • Assumed cost: 0.0406/request (43,644 x 0.75/1M + 2,104 x \$3.75 / 1M)

Business Impact

- Faster handling of routine complaints
- Less manual investigation for support teams
- Clear and auditable decisions

MY APPROACH

1. Understand the business problem and workflow
2. Identify the required data, policies, and outcomes
3. Compare single-agent and multi-agent designs
4. Design specialized agents with shared state
5. Add deterministic guardrails and human approval
6. Use observability and evaluations to improve continuously

CHALLENGE

- **Problem**
 - Too many cases ended in human investigation
- **What tracing revealed**
 - Model API timeouts and rate limits caused incomplete agent outputs
 - Some deterministic checks were overly strict
 - Even a single intermediate failure could escalate the entire case
- **How I addressed it**
 - Added retries for transient model/API failures
 - Introduced a backup model/provider
 - Refined deterministic checks to reduce false escalations
 - Preserved strict approval gates for irreversible actions

FUTURE SCOPE

- **Progressive autonomy:** Automate more low-risk actions after meeting evaluation thresholds.
- **Customer follow-ups:** Add multi-turn conversations to collect missing information.
- **Continuous improvement:** Use production feedback to expand and strengthen the evaluation dataset.

QUESTIONS